Kexuan (Michael) Huang

+1 (608) 320-1127 | +86 15800962172 | kxuan.huang@gmail.com
kx-huang.github.io
Senior Student

EDUCATION

University of Michigan 2022.09 - 2024.08
M.S. in Information Ann Arbor, MI, U.S.

Shanghai Jiao Tong University 2018.09 - 2022.08
B.S.E in Electrical and Computer Engineering (Major GPA: 3.7 / 4.0) Shanghai, China
Minor in Entrepreneurship (GPA: 3.8 / 4.0)

University of Wisconsin-Madison 2022.01 - 2022.05
Exchange student in Computer Science Madison, WI, U.S.

PROFESSIONAL EXPERIENCE

UM-SJTU Joint Institute 2021.05 - 2021.12
Teaching Assistant Shanghai, China

- Hold workshops and Q&A sessions, grade assignments and exams to help nearly 100 students in courses VE300 & VE496.

UM-SJTU Joint Institute 2021.05 - 2021.08
Research Intern Shanghai, China

- Reproduce, simulate and optimize high-speed signal-free intersection models in *SUMO* software with *Python*.

Advanced Micro Devices, Inc. (AMD) 2020.12 - 2021.05
Design Verification Engineer Intern, dGPU SOC, Shanghai R&D Center Shanghai, China

- Work on *HDL* code and waveform simulation to do RTL troubleshooting.
- Use *Python*, *Shell*, *Perl* and *Ruby* to code automation scripts and maintain regression flow for team's efficient debugging. Use *Pandas*, *NumPy*, and *Sklearn* to analyze data, use *Requests and RegEx* to scrape website, and use *Django* to set up webpage.

Segway-Ninebot, Inc. 2020.01 - 2020.03
Embedded Software Engineer Intern, IoT R&D Department. Shanghai, China

- Work on development of autonomous delivery robot, typically in communication & decision-making strategy optimization.
- Use *C* to do embedded programming with *STM* chip, prototype PCB verification and feedback reporting.

PROJECT EXPERIENCE

The Noisy Town, Developer 2021.09 - 2021.12
Tencent Game Development Project (Silver Award)

- Use *Roblox* Game Engine to develop an open world, task-driven, role-playing simulation game, especially for children around 6~12 years old to explore professional interests through game-based learning.

Intelligent firefighting system based on Internet-of-Things technology, Frontend Team Leader 2021.07 - 2021.12
Capstone Project, UM-SJTU Joint Institute

- Cooperate with Shanghai and Shandong Fire Brigade officially. Develop an integrated IoT firefighting system to perform fire risk management with greater accuracy, better efficiency and lower cost. System is deployed on official server for trial run currently.
- Implement backend with *FastAPI,* webpage with *React.js* and a cross-platform app with *React Native*. Use MongoDB as database, whose current data-handling quantity is at 7 million.

Digital Innovation of the Fever Thermometer, Team Leader 2020.09 - 2021.03
Participate in Research Program (Excellent Project: 3 in 26), Antai College of Economics & Management

- Develop a *WeChat* mini-program with *JavaScript, HTML* and *CSS*. Design the circuit and PCB of infrared thermometer, which can connect to mobiles by *Bluetooth*. Temperature registration can be done with greater efficiency in COVID-19 pandemic.

HONORS & AWARDS

2020-2021 Merit Undergraduate Scholarship, SJTU 2021.11
Advanced Teaching Assistant, UM-SJTU Joint Institute 2021.08
2019-2020 Most Improved Scholarship, SJTU 2020.10

SKILLS

- **Languages:** English (IELTS 7.5 / CET-6), Chinese (native)
- **Programming:** Assembly, Verilog, C (Linux), C++, Python, MATLAB, TypeScript, HTML, Shell, SQL, Markdown, LaTeX
- **Development Tools:** Git, GitHub, Drone CI, Vim, VS Code, macOS, Linux
- **Hobbies:** Badminton (former professional athlete, former SJTU badminton team member), Anime Piano